COURIER MANAGEMENT SYSTEM

3. PROJECT DESIGN PHASE

3.1 Problem-Solution Fit

The Courier Management System is designed to solve the common problems faced in traditional courier management. In a manual courier system, customer details, shipment information, delivery status, and payment details are often maintained using paper records or separate systems. This can lead to data duplication, loss of information, delays, and difficulty in tracking shipments.

The proposed system provides a centralized platform for managing the complete courier process, starting from booking a shipment to final delivery. Customers can provide shipment details, while staff can manage customers, shipments, delivery updates, and invoices efficiently.

The system addresses the following problems:

- Difficulty in maintaining courier records manually.
- Lack of real-time shipment status.
- Errors in customer and shipment information.
- Difficulty in tracking delivery progress.
- Time-consuming manual data entry.
- Difficulty in managing delivery agents.
- Problems in maintaining payment and invoice records.
- Lack of centralized information.

The proposed system provides a suitable solution by storing all courier-related information in a structured database. Authorized users can access and update information according to their roles.

3.2 Proposed Solution

Aspect	Design
Customer Management	Customer details such as name, contact number, email, and address can be stored and managed.

Shipment Management	Shipment information such as tracking number, sender, receiver, package type, weight, and destination can be recorded.
Delivery Tracking	The status of a courier can be updated at different stages such as Booked, Picked Up, In Transit, Out for Delivery, and Delivered.
Delivery Agent Management	Delivery agents can be assigned to shipments and can update delivery information.
Invoice Management	Courier charges and payment-related information can be maintained through invoices.
Status Updates	Users can view the current status of shipments and track their progress.

3.3 Four-Layer Architecture

Layer	Components	Example
1. Presentation Layer	This is the user interface layer through which users interact with the system. It provides screens and forms for entering and viewing information.	Login page Customer details page Shipment registration page Delivery update page Invoice page Shipment tracking page
2. Application/Business Logic Layer	This layer contains the main business rules of the system. It processes user requests and performs operations based on the requirements.	Creating a new shipment Assigning a delivery agent Updating shipment status Calculating courier charges Validating customer information
3. Data/Database Layer	This layer is responsible for storing and retrieving system data. It maintains information in an organized manner.	Customer records Shipment records Delivery updates Delivery agent details Invoice details
4. Integration/Service Layer	This layer supports communication between the system and external services or components when required.	Notification services Payment services Email/SMS services External tracking services

3.4 Object and Relationship Design

Object and relationship design defines the major objects in the Courier Management System and the relationship between them.

Main Objects

1. Customer

Stores customer information.

Attributes:

Customer ID

Customer Name

Phone Number

Email

Address

2. Shipment

Stores information about a courier shipment.

Attributes:

Shipment ID

Tracking Number

Sender

Receiver

Source

Destination

Weight

Shipment Status

3. Delivery Agent

Stores details of the person responsible for delivering the shipment.

Attributes:

Agent ID

Agent Name

Phone Number

Availability

Assigned Shipment

4. Delivery Update

Stores the progress of a shipment.

Attributes:

Update ID

Shipment ID

Status

Update Date

Location

Remarks

5. Invoice

Stores payment and courier charge information.

Attributes:

Invoice ID

Shipment ID

Amount

Payment Status

Payment Date

Relationships

3.5 Design Goals

The main design goals of the Courier Management System are:

1. Simplicity

The system should provide a simple and user-friendly interface so that users can easily perform operations.

2. Accuracy

The system should reduce errors during data entry and maintain accurate customer and shipment information.

3. Efficiency

The system should reduce manual work and improve the speed of courier management activities.

4. Security

Only authorized users should be allowed to access or modify sensitive information.

5. Scalability

The system should be designed so that new features and additional users can be added in the future.

6. Reliability

The system should maintain consistent courier records and provide reliable shipment information.

7. Easy Tracking

Users should be able to identify the current status of a shipment using its tracking number.

8. Maintainability

The system should have a well-organized structure so that future modifications and maintenance can be performed easily.

3.6 Expected Outcome

The expected outcome of the Courier Management System is to provide an efficient, organized, and reliable platform for managing courier operations.

After implementing the system:

Customer information can be maintained systematically.

Shipment details can be stored securely.

Courier status can be updated easily.

Delivery agents can be assigned efficiently.

Customers can track shipment progress.

Invoice and payment details can be maintained.

Manual paperwork can be reduced.

Data duplication and errors can be minimized.

Staff can access required information quickly.

Courier management becomes faster and more organized.

Overall, the system is expected to improve the efficiency, accuracy, transparency, and reliability of courier operations. It provides a structured approach to managing customers, shipments, delivery agents, delivery updates, and invoices in a single system.